Experiment 7

Student Name: DIWANSH
UID: 24BAI70294
Branch: CSE - AIML
Section/Group: 24AIT_KRG 1B
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Q1.
Consider a relation R having attributes as R(ABCD), functional dependencies are given below:

AB->C, C->D, D->A

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Solution:
Given,
AB-C, C-D,D->A

B never appears on RHS so B must be a part of candidate keys

Finding closures,
B+= B
AB+ = ABCD
BC+ = BCDA
BD+ = BDAC

Candidate Keys: {AB, BC, BD}
Prime Attributes: {A, B, C, D}
Non-Prime Attributes: {

Checking FDs for BCNF,
AB -> C (AB is a super key)
C->D (C is not a super key so it violates BCNF property)

And since all attributes are prime attributes, relation is in 3NF.

Q2. Relation R(ABCDE) having functional dependencies as:
A->D, B->A, BC->D, AC->BE

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Solution:
Given,
A-D
B->A
BC->D
AC->BE

C never appears on RHS so C must be a part of the candidate key

Finding closures:
C+ = C
AC+= ACBED
BC+ = BCADE
CD+ = CD
CE+ = CE

Candidate Keys: {AC, BC}
Prime Attributes: {A, B, C}
Non-Prime Attributes {D, E}

Checking FDs for BCNF,
A - D (A is not a super key, hence violates property for BCNF)

Checking FDs for 3NF,
A -> D (Neither A is super key nor is D a prime attribute, hence violates 3NF)

Checking FDs for 2NF,
A - D (A is a proper subset of candidate key AC and D is a non-prime attribute, hence violates 2NF)

So, it is in 1NF.

Q3. Consider a relation R having attributes as R(ABCDE), functional dependencies are given below:

B->A, A->C, BC->D, AC->BE

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Solution:
Given,
B->A
A-C
BC ->D
AC-BE

A+ = ACBED
B+ = BACED

Candidate Keys: {A,B}
Prime Attributes: {A, B}
Non-Prime Attributes: {C, D, E}

Checking FDs for BCNF,
B -> A (B is a super key)
A - C (A is a super key)
BC - D (BC is a super key)
AC- BE (AC is a super key)

So the table is in BCNF.

Q4. Consider a relation R having attributes as R(ABCDEF), functional dependencies are given below:

A->BCD, BC->DE, B->D, D->A

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Solution:

Given,

A-BCD
BC-DE
B-D
D->A

F never appears on RHS, so F must be a part of the candidate key

Finding closures,
F+=F
AF+ = ABCDEF
BF+ = BFDACE
CF+ = CF
DF+ =DFABCE
EF+ = EF

Candidate Keys: {AF, BF, DF}
Prime Attributes: {A, B, D, F}
Non-Prime Attributes: {C, E}

Checking FDs for BCNF,
A -> BCD (A is not a super key)

Checking FDs for 3NF,
A -> BCD (A is not a super key and neither are C is non-prime)

Checking FDs for 2NF,
A - BCD (A is a subset of candidate key AF)

So highest NF is 1NF.

Q5. Relation R(ABCDE) having dependencies as:

CE -> D
D ->B
C -> A

Highest normal form? C.K set?

Solution:
Given,
CE->D
D->B

C->A

C and E never appear on RHS so CE must be a part of candidate key

Finding closures,

CE+= CEDBA

Candidate Keys: {CE}
Prime Attributes: {C, E}
Non-Prime Attributes: {A, B, D}

Checking BCNF,
CE->D (CE is a super key)
D-> B (D is not a super key)

Checking 2NF
C-> A (C is a proper subset of candidate key CE, and A is non-prime, hence violates
2NF)

Highest
est Normal Form: 1NF

Q6. Relation R(ABCDEF) having dependencies as:
AB -> C
DC -> AE
E -> F
Highest normal form? C.K set?

Solution:
Given,

AB->C
DC ->AE
E-F

B and D never appear on RHS, so both B and D must be part of every candidate key.

Finding closures,

BD+=BD
ABD+= ABCDEF

BCD+ = ABCDEF

Candidate Keys: {ABD, BCD}
Prime Attributes: {A, B, C, D}
Non-Prime Attributes: {E, F}

Checking BCNF,
AB -> C (AB is not a super key)

Checking 3NF,
DC-> AE (DC is not a super key and E is non-prime)

Checking 2NF,
DC-> E (DC subset of candidate key BCD)

Highest NF: 1NF

Q7. Relation R(ABCDEFGHI) having dependencies as:
AB -> C
BD -> EF
AD -> GH
A -> I
Highest normal form? C.K set?

Solution:
Given,

AB->C
BD ->EF
AD-> GH
A->I

A, B, D never appear on RHS, so A, B and D must be part of every candidate key

ABD+= ABCDEFGHI

Candidate Keys: {ABD}
Prime Attributes: {A, B, D}
Non-Prime Attributes: {C, E, F, G, H, I;

Checking BCNF,

AB -> C (AB is not a super key)

Checking 3NF,
AB-> C (AB is not a super key and C is non-prime)

Checking 2NF,
AB-> C (AB is a proper subset of candidate key ABD and C is non-prime)

Highest NF: 1NF

Q8. Relation R(ABCDE) having dependencies as:
AB -> CD
D -> A
BC -> DE
Highest normal form? C.K set?

Solution:
Given,
AB ->CD
D->A
BC->DE

Since B never appears on RHS, B must be part of every candidate key.

Finding closures,

B+= B
AB+= ABCDE
BD+= ABCDE
BC+= ABCDE

Candidate Keys: {AB, BD, BC}
Prime Attributes: {A, B, C, D}
Non-Prime Attributes: {E}

Checking BCNF,
D-> A (D is not a super key)

Checking 3NF,
D-> A (D is not a super key, but A is prime)

Highest NF: 3NF

Q9. Relation R(ABCDE) having dependencies as:
BC -> ADE
D -> B
Highest normal form? C.K set?

Solution:
Given,
BC - ADE
D->B

C, and E never appear on RHS, so C and E must be part of every candidate key.

Finding closures,
C+=C
BC+= ABCDE
CD+= ABCDE

Candidate Keys: {BC, CD}

Prime Attributes: {B, C, D}
Non-Prime Attributes: {A, E}

Checking BCNF,
D-> B (D is not a super key)

Checking 3NF,
D-> B (D is not a super key, but B is prime)
BC - ADE (BC is a super key)

Highest NF: 3NF

Q10. Relation R(ABC) having dependencies as:
A -> B
B -> C
C -> A
Highest normal form? C.K set?

Solution:
Given,

A->B
B->C
C-A

Finding closures,
A+= ABC
B+= ABC
C+= ABC

Candidate Keys: {A, B, C}
Prime Attributes: {A, B, C}
Non-Prime Attributes: {

All are determinants are super keys so BCNF

Highest NF: BCNF

Q11. Consider a relation P having fields as F1, F2, F3, F4, F5 with the functional dependencies as follows:

F1 -> F3
F2 -> F4
(F1,F2) -> F5

In terms of normalization, this table is in which normal form. Describe the prime and non-prime attributes for the relation P.

Solution:
Given,
F1-F3
F2->F4
(F1,F2)-F5

F1 and F2 never appear on RHS, so they must be part of every candidate key.

Finding closures,
F1F2+=F1F2F3F4F5

Candidate Key: {F1F2}
Prime Attributes: {F1,F2}
Non-Prime Attributes: {F3, F4,F5}

Checking BCNF,
F1->F3 (F1 is not a super key)

Checking 3NF,
F1->F3 (F1 is not a super key and F3 is non-prime)

Checking 2NF,
F1->F3 (F1 is a proper subset of candidate key F1F2 and F3 is non-prime)

Highest NF: 1NF

Q12. Consider a relation R(ABCDEFGHIJ) having FDs as:

AB -> C
AD -> GH
BD -> EF
A -> I
H -> J

Identify that relation is in which normal form and the candidate key by determining the prime and non-prime attributes.
Apply the decomposition if needed to remove the redundancy.
Identify how many new tables will be formed after the decomposition.

Solution:
Given,
AB->C
AD -> GH
BD ->EF
A-I
H->J

A, B and D never appear on RHS, so A, B and D must be part of every candidate key

Finding closures,
ABD+ = ABCDEFGHIJ

Candidate Key: {ABD}
Prime Attributes: {A, B, D}
Non-Prime Attributes: {C, E, F, G, H, I, J}

Checking BCNF,
AB -> C (AB is not a super key, so relation violates BCNF)

Checking 3NF,
AB-> C (AB is not a super key and C is non-prime, so violates 3NF)

Checking 2NF,
AB -> C (AB is a proper subset of candidate key ABD and C is non-prime, so it violates 2NF)

Highest NF: 1NF

Decomposition to remove redundancy

Since the relation violates 2NF, decompose the partial dependencies:

R1(A, B, C) for AB->C

R2(A, D, G, H) for AD-GH

R3(B, D, E, F) for BD -EF

R4(A, I) for A ->I

We also need a relation containing the candidate key:

R5(A, B, D) for candidate key

Since H-> J is a transitive dependency, if we also want to remove that redundancy and reach 3NF, create:

R6(H, J) for H-> J

So, 6 tables are formed.

Q13. Relation R having attributes as (A, B, C, D, E) having the functional dependencies as follows:

A -> B
B -> E

C -> D

Identify that relation is in which normal form and the candidate key by determining the prime and non-prime attributes.
Apply the decomposition if needed to remove the redundancy.
Identify how many new tables will be formed after the decomposition.

Solution:
Given,

A-B
B-E
C->D

A and C never appear on RHS, so A and C must be part of every candidate key.

Finding closures,
AC+= ABCDE

Candidate Key: {AC}
Prime Attributes: {A, C}
Non-Prime Attributes: {B, D, E}

Checking BCNF,
A -> B (A is not a super key, so it violates BCNF)

Checking 3NF,
A - B (A is not a super key and B is non-prime, so it violates 3NF)

Checking 2NF,
A -> B (A is a proper subset of candidate key AC and B is non-prime, so it violates 2NF)

Highest NF: 1NF

Decomposition
To remove the partial and transitive dependencies, decompose as:

R1(A, B) for A->B

R2(B, E) for B->E

R3(C, D) for C->D

R4(A, C) for Candidate key AC

Number of new tables formed:

4 tables